

# Vishwas Kothari

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M.S. Computer Science candidate graduating May 2027 and available June 2027. Builds machine-learning and retrieval systems in Python, and the evaluation frameworks that establish whether they actually work. Interested in the engineering around models rather than models alone: guardrails on what a system is allowed to do, measurements that survive scrutiny, and outputs that non-specialists can inspect and act on.

## EDUCATION

### University of Colorado Boulder

Professional Master's in Computer Science — GPA: 3.74/4.0

Aug 2025 – May 2027

Boulder, CO

Coursework: Neural Networks & Deep Learning, Theory of Machine Learning, Design & Analysis of Algorithms, Linear Programming, Database Systems

Database Systems coursework: processed one million records with PySpark and Spark SQL on a standalone Spark cluster, and cut repeat-query latency from 176 ms to 20 ms with a Redis cache over a 500,000-row PostgreSQL table behind a Node.js service.

### Vellore Institute of Technology

B.Tech in Information Technology — GPA: 8.49/10.0

Aug 2021 – Jun 2025

Vellore, India

## PROJECTS

### FinRAG Auditor — Retrieval and Agentic Evaluation over Financial Reports

GitHub

Python, LangGraph, LangChain, BM25, BGE embeddings, cross-encoder reranking, pytest, GitHub Actions, Docker

- **Built a hybrid retrieval pipeline** fusing BM25 with dense embeddings through reciprocal rank fusion and cross-encoder reranking, evaluated with Recall@K, MRR and nDCG across 883 questions over an 8,783-chunk corpus.
- **Implemented bounded, auditable tool execution** in a LangGraph workflow: each generated plan is validated before anything runs, plans whose citations or numeric operands are absent from retrieved evidence are rejected, arithmetic is restricted to a safe Decimal evaluator, and weak evidence routes to explicit abstention.
- **Selected components through controlled evaluation:** replacing TinyBERT with BGE improved MRR by 10.92 points across 883 questions (95% bootstrap CI: 8.65–13.25), while an oracle-evidence audit improved execution accuracy by only 3.96 points, isolating planning as the larger bottleneck. Shipped with 85 tests at 93% line coverage and a Docker image built and smoke-tested in CI.

### FinAgent-Eval — Evaluation Framework for Document AI and Vision-Language Models

GitHub

Python, PyTorch, Hugging Face Transformers, pytest, GitHub Actions, Streamlit

- **Built an evaluation framework benchmarking five models** on 397 hand-curated questions across 79 financial-document images from 10 public companies, with unit-aware grading, bootstrap confidence intervals, a 13-label failure taxonomy and a seven-tab dashboard.
- **Audited the evaluation layer and found the grader itself was wrong:** 35 of 151 v1 failures were correct answers marked down because the grader could not parse financial unit words. Rebuilt and versioned it, regenerated every metric, and expanded the suite from 12 to 96 tests with reproducibility CI that detects artifact and metric drift.
- **Exposed failures hidden by aggregate performance:** the leading model reached 68.77% exact match and 0.7378 ANLS overall, yet answered only 10 of 65 numerical-reasoning questions correctly; none of six numerical tasks remained answer-consistent across four phrasings.

### XAI Credit Lens — Credit-Risk Modeling and Model Governance

GitHub

Python, Scikit-learn, XGBoost, LightGBM, Optuna, SHAP, LIME, DiCE, Streamlit

- **Modeled consumer-credit default across 30,000 customer records**, comparing three model families under stratified five-fold cross-validation with Optuna tuning, and evaluating discrimination and calibration separately at 0.782 test ROC AUC and a 0.180 Brier score.
- **Audited outcomes across four protected attributes** and shipped a review application exposing attributions, counterfactual queries and governance panels mapped to ECOA and Federal Reserve SR 11-7, so a non-technical reviewer could interrogate an individual decision without running code.

## EXPERIENCE

### Space Applications Centre — Indian Space Research Organisation (ISRO)

Ahmedabad, India

Research Intern, Biological Oceanography Division

Jan 2025 – Jul 2025

- **Debugged convergence failures across inputs spanning three orders of magnitude** while implementing and comparing a Levenberg–Marquardt framework and a custom Particle Swarm Optimization workflow, resolving them through log-space reparameterization.
- **Validated the models against observed data** using RMSE, MAPD, relative bias and  $R^2$ , reaching  $R^2 = 0.987$  on reflectance reconstruction and log-space  $R^2 = 0.879$  across 105 chlorophyll validation samples.
- **Automated a manual multi-source data workflow** in Python on Linux for OCM and MODIS Aqua products supporting OCM-3 mission validation, and translated the results for multidisciplinary researchers through six diagnostic visualization types and a 44-page report presented across divisions.

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, MATLAB, Java, C, C++, JavaScript

**AI & LLM Systems:** RAG, agentic workflows, LangGraph, LangChain, prompt engineering, guardrails, quantized inference

**Retrieval & NLP:** BM25, dense embeddings, hybrid retrieval (RRF), cross-encoder reranking, Recall@K, MRR, nDCG

**ML & Deep Learning:** PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, LightGBM, Optuna, LoRA/PEFT

**Statistics & Evaluation:** Cross-validation, bootstrap intervals, calibration, ablation design, failure taxonomies

**Data & Systems:** PostgreSQL, Redis, Apache Spark, PySpark, Spark SQL, Pandas, NumPy, SciPy, Linux

**Engineering:** Git, Docker, GitHub Actions (CI), pytest, Ruff, mypy, coverage enforcement, reproducible pipelines

## PUBLICATION, PEER REVIEW AND CERTIFICATION

**Publication:** Kothari, V. and Krishwanth, B. (2025). *Empowering Survivors: Ethical Artificial Intelligence for Countering Violence Against Women (VAW)*. Springer.

**Peer Reviewer:** *Public Health; Social Sciences & Humanities Open* (Elsevier, 2026–Present).

**Certification:** AWS Certified Cloud Practitioner (2024).